## Study design: trace each failure, then replay the moment it was decided

Tracing of 104 failed outputs, controlled replays at 13 failure points, and direct probes of the same facts

### 16 benchmark runs

13 BixBench3 task runs
(GPT-6.1 Sol root agent,
DeepSeek V4.1 Flash sub-agents)
3 GeneBench-Pro runs of 10 problems
(DeepSeek V4.1 Flash ×2, GPT-6 Astra)

### 104 failed outputs, traced

each reproduced against the reference
smallest change that removes it
transcript read back to the step
where the error entered
two independent passes, κ = 0.79

### Where failures occur

primary cause per output
48 cause groups
printed disproving evidence:
42 of 58 GeneBench-Pro keys
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### 13 failure points

the exact model request sent just
after the disproving number was
printed; 3 points from passing
runs as controls

### Controlled replays

Claude Haiku 5.5 answers in the
original agent's place
11 conditions × 6 samples
scored by Claude Sonnet 5.5
second-judge agreement κ = 0.73

### Direct probes

the fact that would have prevented
a failure, asked outside the task

5 probes, 3 to 30 samples each
compared with what the agent did
in the run

Section 5